Lab Manual: Site Suitability Analysis using QGIS

Course: Advanced Spatial Analysis & GIS

Topic: Multi-Criteria Evaluation (MCE) for Site Selection using QGIS

1. Overview and Learning Objectives

Site suitability analysis is a critical application of Geographic Information Systems (GIS) used to identify optimal locations for development, conservation, or facility planning. This lab manual walks through the process of combining spatial factors using continuous raster processing, reclassification, and weighted overlay analysis.

By completing this laboratory exercise, students will be able to:

- Convert vector features into Euclidean distance (proximity) rasters.
- Calculate feature concentration using line density estimation tools.
- Standardize spatial resolution and map extent across multiple raster layers.
- Normalize continuous spatial data onto a unified evaluation scale using raster reclassification.
- Execute a multi-criteria Weighted Overlay Analysis using the QGIS Raster Calculator.

2. Methodological Workflow Summary

Workflow Stage	GIS Operation	Primary QGIS Tool	Output Dataset
Proximity Analysis	Vector to Raster Conversion & Distance	Rasterize / Proximity (Raster Distance)	proximity_water.tif
Density Analysis	Kernel/Line Density Computation	Line Density (Processing Toolbox)	road_density.tif
Data Standardization	Spatial Resampling /	Warp (Reproject) /	resampled_slope.tif

Workflow Stage	GIS Operation	Primary QGIS Tool	Output Dataset
	Align Extents	Align Rasters	
Criteria Normalization	Reclassification by Table	Reclassify by Table	reclass_*.tif
Suitability Overlay	Weighted Raster Addition	Raster Calculator	site_suitability_final.tif

3. Step-by-Step Practical Instructions

Step 1: Generate Proximity Maps (Euclidean Distance)

Proximity maps compute the shortest straight-line distance from every cell in a raster to the nearest feature of interest (e.g., water bodies or infrastructure).

1. Load your feature vector layer (e.g., water_bodies.shp) into QGIS via **Layer > Add Layer > Add Vector Layer**.
2. Rasterize the vector feature:
   - Navigate to **Raster > Conversion > Rasterize (Vector to Raster)**.
   - Set **Input layer** to water_bodies.
   - Set **Field to use for burn-in value** to a constant value of 1.
   - Set **Output raster units** to *Georeferenced units* and set Resolution to 30 meters.
   - Save output as water_binary.tif.
3. Compute Euclidean distance:
   - Navigate to **Raster > Analysis > Proximity (Raster Distance)**.
   - Set **Input layer** to water_binary.tif.
   - Set **Distance units** to *Georeferenced units*.
   - Save output as proximity_water.tif.

Step 2: Compute Line Density

Line density calculates the concentration of linear features per unit area within a specified search neighborhood (e.g., road network access or drainage density).

1. Load linear feature layer (e.g., roads.shp).
2. Open the **Processing Toolbox** (Ctrl + Alt + T).
3. Search for and open **Line density**.
4. Configure the processing parameters:

- **Input layer:** roads.shp
- **Radius / Search Distance:** Set search neighborhood (e.g., 1000 meters).
- **Pixel Size:** Set target cell size to 30 meters.

5. Save the resulting dataset as road_density.tif.

Step 3: Resample and Align Rasters

Prior to performing raster math, all layers must match in spatial resolution (cell size), Coordinate Reference System (CRS), and geographic extent.

1. Open **Processing Toolbox > Raster Analysis > Align Rasters** (or **Raster > Extraction > Warp (Reproject)** for individual files).
2. Using **Warp (Reproject)**:
   - **Input layer:** Select target raster needing alignment (e.g., DEM or land cover).
   - **Target CRS:** Set project projected coordinate system (e.g., UTM Zone 44N).
   - **Resampling method:** Use Bilinear or Cubic for continuous data (e.g., elevation/slope), and Nearest Neighbor for categorical datasets.
   - **Output cell size:** Set explicitly to 30.0.
3. Save adjusted rasters to project directory.

Step 4: Reclassify Criteria Datasets

To combine datasets with different measurement units (meters, percent, density values), reclassify all inputs into a common scale (e.g., 1 = Least Suitable, 5 = Most Suitable).

1. Open **Processing Toolbox > Raster Analysis > Reclassify by Table**.
2. Set **Raster layer** to proximity_water.tif.
3. Click **Reclassification table [...]** and define value breaks:

Minimum Range (m)	Maximum Range (m)	Suitability Score (1-5)
0	500	1 (Unsuitable Buffer Zone)
500	2000	5 (Highly Suitable)
2000	5000	3 (Moderately Suitable)
5000	10000	1 (Poor Accessibility)

4. Set **Boundaries treatment** to *min* < *value* <= *max*.
5. Save output as reclass_water.tif. Repeat this procedure for all remaining criteria layers (e.g.,

reclass_density.tif, reclass_slope.tif).

Step 5: Execute Weighted Overlay Analysis

Weighted Overlay combines all normalized factors by multiplying each layer score by its designated criteria percentage weight.

Evaluation Model Weights:

- Water Proximity (Weight: 30% / 0.30)
- Road Density (Weight: 40% / 0.40)
- Terrain Slope (Weight: 30% / 0.30)

1. Open **Raster > Raster Calculator**.
2. Build the multi-criteria equation in the expression window:
   `("reclass_water@1" * 0.30) + ("reclass_density@1" * 0.40) + ("reclass_slope@1" * 0.30)`
3. Set **Output layer** to site_suitability_final.tif.
4. Set **Output Format** to GeoTIFF (GTiff) and execute the calculation.

4. Final Map Styling & Verification

Once generated, style site_suitability_final.tif to present results clearly:

1. Right-click layer > **Properties > Symbology**.
2. Set **Render type** to *Singleband pseudocolor*.
3. Select a standard **Red-Yellow-Green** color ramp (where Green represents score 5 / High Suitability).
4. Select classification mode as *Equal Interval* or *Discrete*.